## Glossary

**Tracks and sectors:** A hard disk consists of platters on which data is stored. These platters consist of tens of thousands of tracks that are tightly integrated in concentric circles. These tracks contain loads of vital information, and thus are broken down into sectors. Hence, a sector is the smallest addressable unit that can hold 512 bytes of information.

**S.M.A.R.T:** The Self-Monitoring Analysis and Reporting Tool system uses internal hard disk monitoring technology to analyse your hard disk, and prevent hard disk failure, or data loss.

**Bad sector:** A sector that cannot be used due to a physical flaw on the disk is called a bad sector.

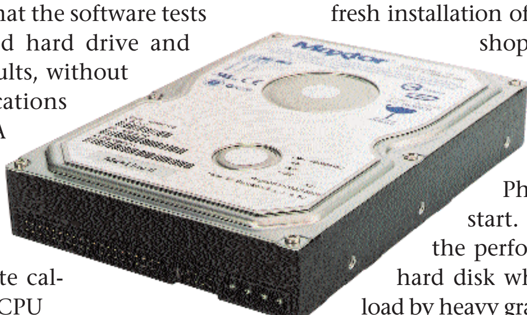
**Clusters:** A cluster is a logical unit of storage on a hard disk, or a floppy drive. The size of a cluster varies from 512 bytes, to 256 kilobytes depending on the size of the partition, and the file system.

**FAT:** The FAT (File Allocation Table) tells the Disk Operating System which portions of the disk belong to each file. The FAT links together all the clusters belonging to each file, no matter where they are on the disk.

**Partition:** By default, whenever a hard disk is formatted, it will have one partition. Hence, a partition is a formatted section of a hard disk. Large hard disks can be formatted into multiple partitions, and each partition acts as an individual, or a separate hard drive.

**Boot disk:** A special floppy boot disk allows your PC to boot, even if it cannot boot from the hard disk. This disk is also known as a System Rescue Disk.

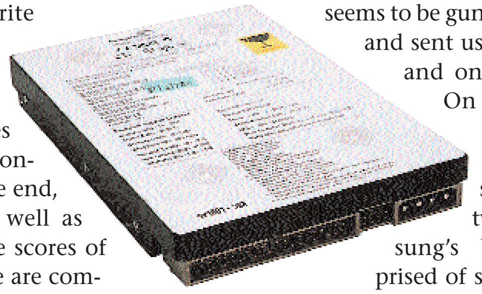
conducted on the hard drives. HDTach 2.6 is used to test the hard drives without installing any other software on it. The logic behind this is that the software tests the raw unformatted hard drive and provides us with results, without making any modifications to the hard drive. A graph displays the results, as the data is read and written in real time. After the test is done, the suite calculates the total CPU power consumed by the drive during the test. In real world conditions, the drive is not likely to excede this CPU utilisation percentage since the disk is continuously accessed during this test.



*Maxtor DiamondMax PlusII 6Y160M0*

### SiSoft Sandra 2003 Pro (File System benchmark)

This is a synthetic benchmark, and gives us vital information about the drive, such as transfer rate, sequential and random read as well as write speed, along with average access time. The benchmarking suite reads and writes the data to the disk continuously, and in the end, gives individual as well as overall scores. These scores of the tested hard drive are compared to that of other hard drives on a reference score sheet.



*Seagate Barracuda ST340015A*

**Real world tests (File transfer tests):** This is a new category that we have introduced to test hard drives. This test mimics a user's daily usage pattern. A user generally moves, or copies data from one place to another. It may just take a few seconds for a single file, but when moving huge amounts of data, it could take anything from a few minutes, to an hour. This can really get frustrating, and hence it becomes important to test the hard drive in a real world environment. The tests consisted of transferring an assorted file set of 1 GB, and a single file of 1 GB from one partition of the hard drive to another.

These test results are very important as they directly reflect the true performance capability of the hard drive's read and write speeds in a real world environment. Poor scores in this test indicate lower read and write speed capabilities of the hard drive.

**Adobe Photoshop load time (Application load time):** In this test, we made a fresh installation of Adobe Photoshop on the hard drive, and noted down the time taken for Photoshop 7 to start. This test gives the performance of the hard disk when put under load by heavy graphics software such as Photoshop.

**Adobe Photoshop file load time:** This is a very stressful test in which we loaded, and recorded the time taken for Adobe Photoshop to open a 200 MB TIFF image file.

### Test analysis

In all we tested 13 hard drives from three major manufacturers—Seagate, Samsung, and Maxtor. Of these 13 drives, Maxtor seems to be gunning for the future, and sent us three SATA drives, and one ultra ATA drive. On the other hand, Seagate seems to be playing safe, and sent one SATA, and two IDE drives. Samsung's bandwagon comprised of six IDE drives, with not a single SATA drive. The drive capacity ranged from a minimum of 40 GB to a mammoth 250 GB. Let's see how the drives fared on both the performance, and the features front.

### Features, specifications, and bundled accessories

On the features front, not a single drive matches the specifications of the Maxtor Maxline Plus II—it simply blows the competitors away. The Maxtor Maxline Plus II comes with a mammoth 250 GB of storage space, and an impressive 8 MB buffer, which makes for an unbeatable performance combination. All these features are further enhanced by the provision of the Ultra ATA 133 interface. In fact, all the tested Maxtor drives came with a 7,200 rpm spindle, and an 8 MB buffer. Not only is a larger buffer good for performance, it also comes in handy during disk intensive tasks.

The Seagate ST3160021A, sported the second best feature list amongst the ATA drives, followed by the twins from Samsung. The Samsung SP0411N, and SP0802N drives came in third, with a good amount of storage space, an ATA 133 interface, and a 7,200 rpm spindle speed, making them an ideal choice for desktop data storage, at an irresistible price. The other ATA hard drives, especially those from Samsung, were not feature rich but yet made an excellent bang for the buck with a decent performance, and price factors taken into consideration.

In the SATA category, nothing came close to the 200 GB Maxtor DiamondMax Plus. It comes with an 8 MB buffer that compliments the 7,200 rpm spindle speed really well. All the SATA drives from Maxtor come with both, the SATA and Molex